CASE STUDY 17

STUDENT PERFORMANCE TREND ANALYSIS

Python Program:

```
print("===== STUDENT PERFORMANCE TREND ANALYSIS =====")


subjects = ["Python", "Java", "Mathematics", "Data Structures"]


marks = []


for subject in subjects:
    mark = float(input("Enter marks in {} (out of 100): ".format(subject)))


    if mark < 0 or mark > 100:
        print("Invalid marks! Enter marks between 0 and 100.")
        break


    marks.append(mark)


if len(marks) == len(subjects):
    total = sum(marks)
    average = total / len(marks)


    print("\n===== PERFORMANCE REPORT =====")
    print("Total Marks:", total, "out of", len(subjects) * 100)
```

```
print("Average Marks: {:.2f}".format(average))

if average >= 90:
    grade = "A+"
elif average >= 80:
    grade = "A"
elif average >= 70:
    grade = "B"
elif average >= 60:
    grade = "C"
elif average >= 50:
    grade = "D"
else:
    grade = "F"

print("Overall Grade:", grade)

highest_mark = max(marks)
lowest_mark = min(marks)

print("Highest Marks:", highest_mark)
print("Lowest Marks:", lowest_mark)

print("\n===== SUBJECT-WISE ANALYSIS =====")
```

```
    for i in range(len(subjects)):
        print("{}: {} marks".format(subjects[i], marks[i]))

    if marks[-1] > marks[0]:
        print("Trend: Performance improved from the first to the last subject.")
    elif marks[-1] < marks[0]:
        print("Trend: Performance decreased from the first to the last subject.")
    else:
        print("Trend: Performance remained the same from the first to the last subject.")

    print("\nAnalysis completed.")
```

Output:

